

Methodology of the Engineering Sciences

Till Grüne-Yanoff

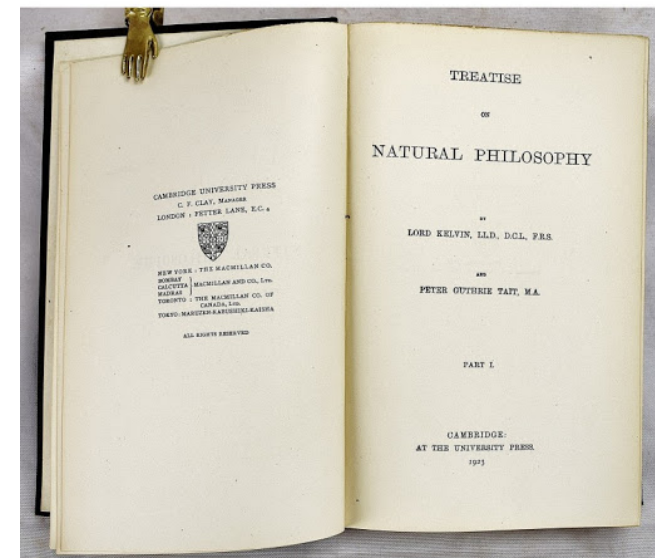
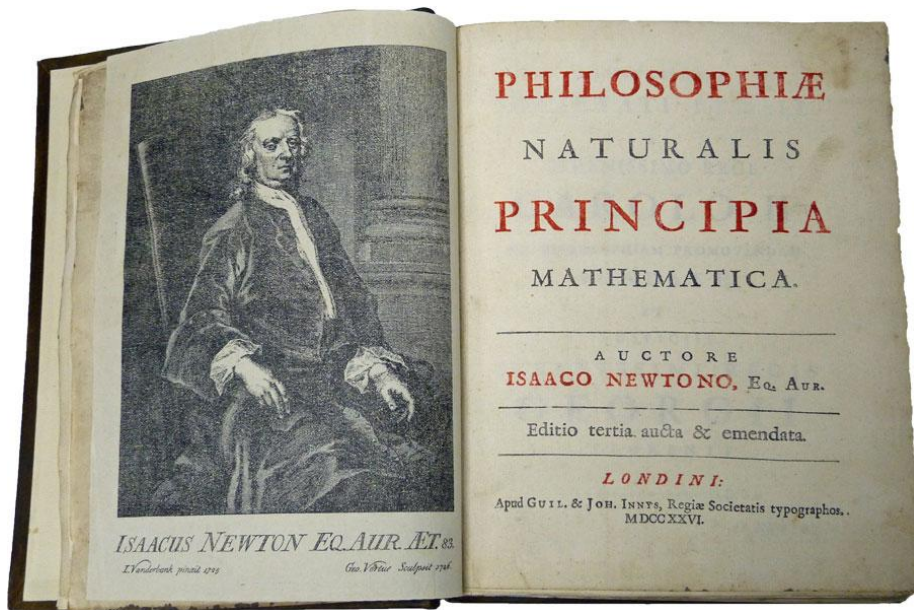
1. What's special about
the engineering sciences?

Disparate origins

- Let me start by sketching in broad strokes the respective historical roots of the natural sciences and of the engineering, and in this course also clarify what I mean by “engineering sciences”.

Disparate origins: Science

- What we now call “natural science” has its roots in what until the 19th century was called
 - “**natural philosophy**”. Pre-Socratics, Aristotle, Galileo, Isaac Newton's *Philosophiæ Naturalis Principia Mathematica* (1687); Lord Kelvin's book *Treatise on Natural Philosophy* (1867).

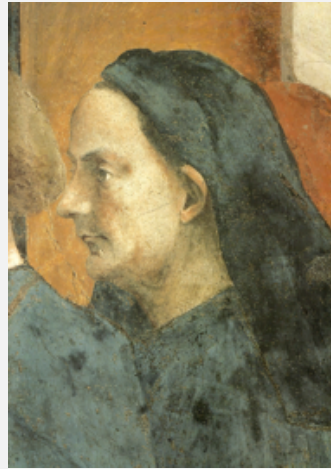


Disparate origins: Engineering

- Those taught in the tradition of natural philosophy were not the ones who build the aqueducts, castles or cathedrals!



Disparate origins: Engineering



Filippo Brunelleschi
(1377- 1446)

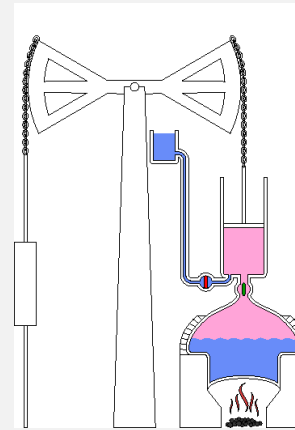
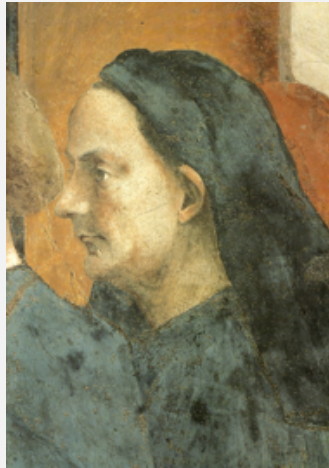
- Trained as a goldsmith
- Worked in sculptor's workshop

Disparate origins: Engineering



Filippo Brunelleschi
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- Trained as a goldsmith
- Worked in sculptor's workshop



Thomas Newcomen
(1664-1729)

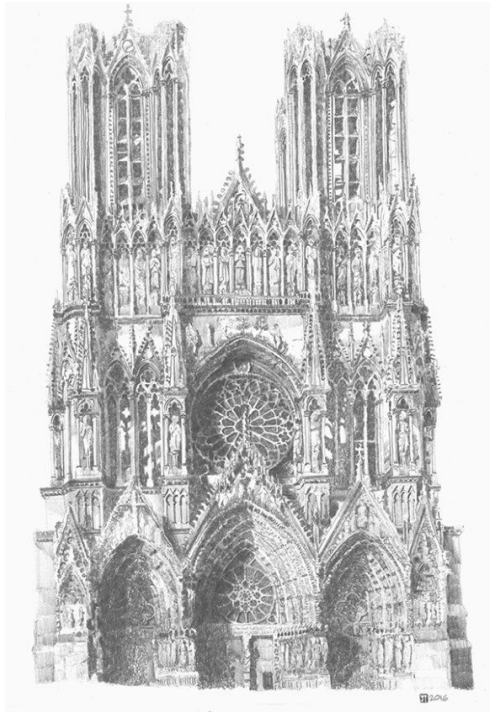
- Inventor of the Newcomen engine
- Worked as an ironmonger (selling tools to mines)

James Watt

Trained as an instrument
maker

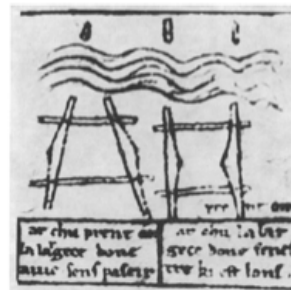
Disparate origins: Engineering

- That these early engineers were craftsmen is particularly clear in the builders of the magnificent gothic cathedrals in France & England 12th - 16th c.



Disparate origins

- Knowledge accumulated by trial and error. No evidence that cathedral builders had formal training in mathematics/geometry or were in fact literate.
- From surviving documents: master masons, if at all, used primitive mathematics, lacking a Euclidian framework. Also highly local, exhibiting "the particular procedures of particular master masons at particular times and places" (Shelby 1972, 421). "organically grown and handed down from individual teacher to student."



. Surveying technique, from Villard de Honnecourt's Sketchbook (ed. Hahnloser, Pl. 39 1, m).

Interim analysis of what I said so far

<u>Sciences</u>	<u>Engineering</u>
aim at understanding	Aim at creating artefacts with particular properties: satisfying clients, efficiency, safety, beauty,...
“explain the world!”	“Change the world!”
Use of abstract reasoning & theory from the start: Standardized theories – , Aristotelean works, math, logic,	Highly specified & local knowledge based on trial and error
Transmitted & taught through books and lecturing	Skills acquired through on the job training by skilled seniors

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} **Tacit knowledge**

Tacit Knowledge

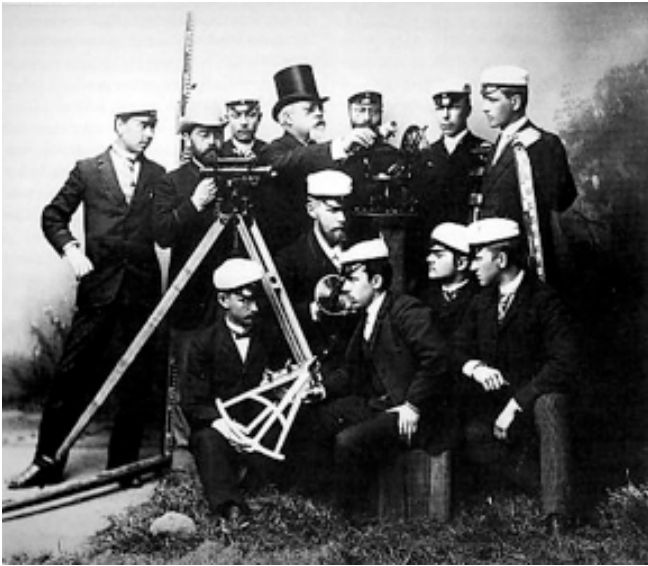
- skills, ideas and experiences that people have but that are not codified and that are difficult to transfer to another person by verbalizing it (Polyani): e.g. riding a bicycle

Modern civil engineering education

- Engineering education began separating itself only about 200 years ago from this close connection to the crafts,
- by turning tacit, local and trial-and-error knowledge **into** explicit, general, and systematic knowledge.

Modern civil engineering education

- By opening centralized training institutions for engineers...



Helsinki University of Technology
student in the late 19th century.

- Ecole Polytechnique, 1794, Paris
- Bauakademie, 1799, Berlin
- Teknologiska Institutet, 1827, Stockholm

Modern civil engineering education

-and eventually also admitting women to these institutions



Helsinki University of Technology
student in the late 19th century



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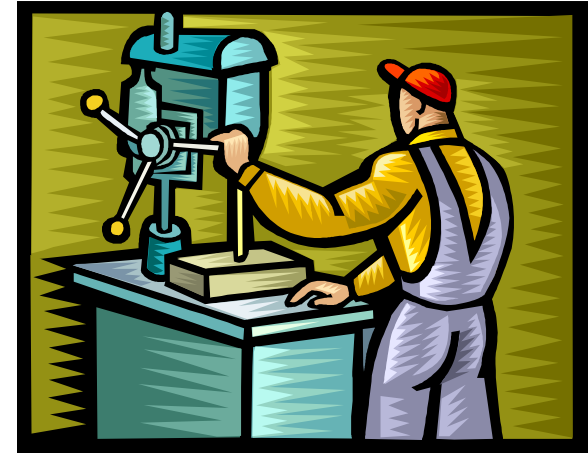
Nora Stanton Blatch Barney (1883 - 1971)

First woman to achieve a degree in engineering. 1906, Cornell University
(first woman to obtain a bachelor's degree in US: 1840)

Modern civil engineering education

....by replacing the workshop with the lecture hall

In 1885, Robert Thurston (Cornell) pushed to reduce “shop” hours and add basic science in the classroom.



The emergence of engineering science

....and by introducing advanced degrees into engineering education

- 1899: Prussian technical institutes confer Dr.-Ing.
- “Technological sciences”, “Engineering sciences”,
“Ingenieurwissenschaften”, “Ingenjörsvetenskap”

Technische Universität Berlin

Summary: What's special about the engin. scie.?

- Goal: design of artefacts (not: their production or maintenance)
- Design requires specialized knowledge bases (more on this later)
- In crafts, this is largely tacit knowledge acquired through trail-and-error and emulating a master

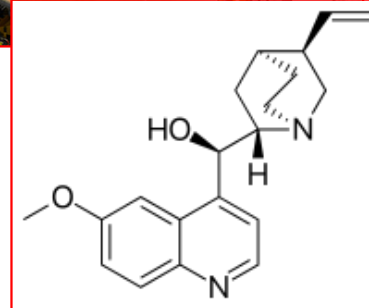
Summary: What's special about the engin. scie.?

- In modern engineering, these knowledge bases are systematically developed
 - Development of dedicated engineering theory
 - Division of epistemic labor – not all engineers design. Many instead contribute to design knowledge
 - Systematic variation of possible solutions
 - Systematic testing under varying conditions
 - Different types of tests, beyond direct trial-and-error

2. Technical artefacts

Examples of technical artefacts

- Objects, processes, systems



What are not technical artefacts?

- Ideas
 - Electrons
 - Human hearts
- } ...not *artefacts*
-
- Representations
 - Social institutions
 - Art objects
- } ...not *technical* artefacts

What distinguishing principles?

1. Artefacts are existing, material objects, not mere ideas or descriptions.

Hence not:

- Ideas

What distinguishing principles?

1. Artefacts are existing, material objects, not mere ideas or descriptions.
2. Artefacts are human-made, not “natural”.

Hence not:

- Electrons
- Human hearts

What distinguishing principles?

1. Artefacts are existing, material objects, not mere ideas or descriptions.
2. Artefacts are human-made, not “natural”.

- *Incidental tools* are often not human-made. Examples: paleolithic hammer stones



What distinguishing principles?

1. Artefacts are existing, material objects, not mere ideas or descriptions.
2. Artefacts are human-made, not “natural”.

- Physicists also make non-natural materials, like e.g. elements 95 to 118, or radioisotopes of naturally occurring elements – but these are not artefacts

Periodic table of the elements

Alkali metals

Alkaline-earth metals

Transition metals

Other metals

Other nonmetals

Halogens

Noble gases

Rare-earth elements (21, 39, 57–71) and lanthanoid elements (57–71 only)

Actinoid elements

period	group 1*																	18	
1	1 H	2																	2 He
2	3 Li	4 Be																	10 Ne
3	11 Na	12 Mg	3	4	5	6	7	8	9	10	11	12	13 Al	14 Si	15 P	16 S	17 Cl	18 Ar	
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr	
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe	
6	55 Cs	56 Ba	57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn	
7	87 Fr	88 Ra	89 Ac	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og	
lanthanoid series 6			58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu			
actinoid series 7			90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr			

*Numbering system adopted by the International Union of Pure and Applied Chemistry (IUPAC). © Encyclopædia Britannica, Inc.

What distinguishing principles?

1. Artefacts are existing, material objects, not mere ideas or descriptions.
2. Artefacts are human-made, not “natural”.

- Physical processes of artefacts and artefact uses are governed by laws of nature – they are “natural” in this way

What distinguishing principles?

1. Artefacts are existing, material objects, not mere ideas or descriptions.

~~2a. Artefacts are human-made, not “natural”.~~

2b. Artefacts are purposeful (they have a *function*), while the identity of non-artefacts does not depend on humans having assigned a purpose to them.

Hence not:

- Electrons

What distinguishing principles?

1. Artefacts are existing, material objects, not mere ideas or descriptions.

~~2a. Artefacts are human-made, not “natural”.~~

2b. Artefacts are identified by their identity of non-artefactuality and are assigned a purpose.

Hence not:

- Electrons
- Human hearts

Interesting case. Biologists identify organs through their functions “anything that pumps blood through the blood vessels of an organism’s circulatory system is a heart”

But: hearts are *ascribed* this function ex post in order to understand organism. But nobody *assigned* them this function: nobody shaped hearts with the intention that they serve this function. They are not designed (unless you believe in intelligent design).

Some organisms, however, are designed – e.g. through breeding or genetic engineering. Thus milk cows, or GMO rapeseed *are* artefacts.

What distinguishing principles?

1. Artefacts are existing, material objects, not mere ideas or descriptions.

~~2a. Artefacts are human-made, not “natural”.~~

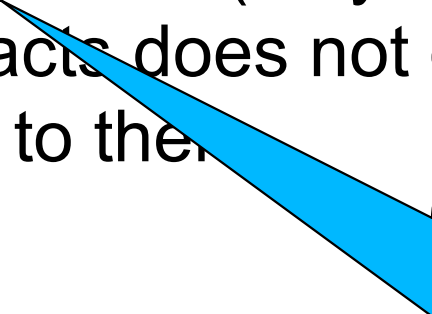
2b. Artefacts are purposeful (they have a *function*), while the identity of non-artefacts does not depend on humans having assigned a purpose to them.

In contrast:

A hammer would not be a hammer (nor a satellite a satellite) without someone describing what she intends to do with it - its purpose/function

What distinguishing principles?

1. Artefacts are existing, material objects, not mere ideas or descriptions.
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Social institutions are art objects are purposeful material objects and hence artefacts. But they are not *technical* artefacts. Why?

What distinguishing principles?

1. Artefacts are existing, material objects, not mere ideas or descriptions.
2. Artefacts are purposeful (they have a *function*), while the identity of non-artefacts does not depend on humans having assigned a purpose to them.
3. Technical artefacts realize their function in virtue of their physical properties.

Hence not:

- Social institutions – realize their function through common acceptance of rules
- Representations/Art objects

What distinguishing principles?

1. Artefacts are existing, material objects, not mere ideas or descriptions.
2. Artefacts are purposeful (they have a *function*), while the identity of non-artefacts does not depend on humans having assigned a purpose to them.
3. Technical artefacts realize their function through their physical properties.

Hence not:

- Social institutions – realize their function through their social properties
- Representations/Art objects

The beautiful paintings of **Lubaina Himid**, which illustrate many of the lecture slides, have their function not through their physical properties but through the symbolic references that give them meaning

The dual nature of artefacts

Technical artefacts are characterized *both* by their **physical properties** *and their intended function*. They serve their functions in virtue of their physical properties

The dual nature of artefacts

Nicely illustrated by Edison's patent application for lightbulb that you have read

intended function

Gives light by incandescence

Offers resistance to electric current

Provides slight surface from which radiation can take place



physical properties

Carbon wire

Platina wires

Vacuum-bulb

Glass receiver

The dual nature of artefacts

Two separate ways of describing technical artefacts: functional and physical.

Both are needed to characterize TAs

- Edison could not fully describe his invention by **either** a **purely** physical or by a **purely** functional description

Neither can be reduced to the other

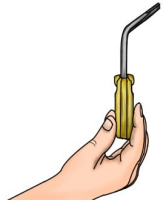
- There is no way to **logically infer** from a functional description of the lightbulb what (all) its physical properties are. Nor can one infer from a purely physical description of the lightbulb what its functions are

But both constrain each other

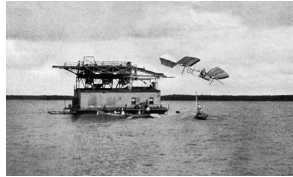
- If a thing had certain physical properties, it could not function as a lightbulb
- For a thing to have certain functions, it has to have certain arrangements of physical properties.

Difficult cases

- Malfunctioning devices



- Failed interventions



Langley's *Aerodrome* in
1903

- Incidental tools



- Misuse

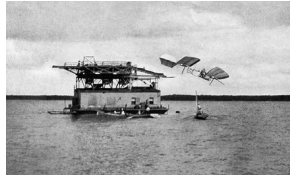


- Obsolete devices



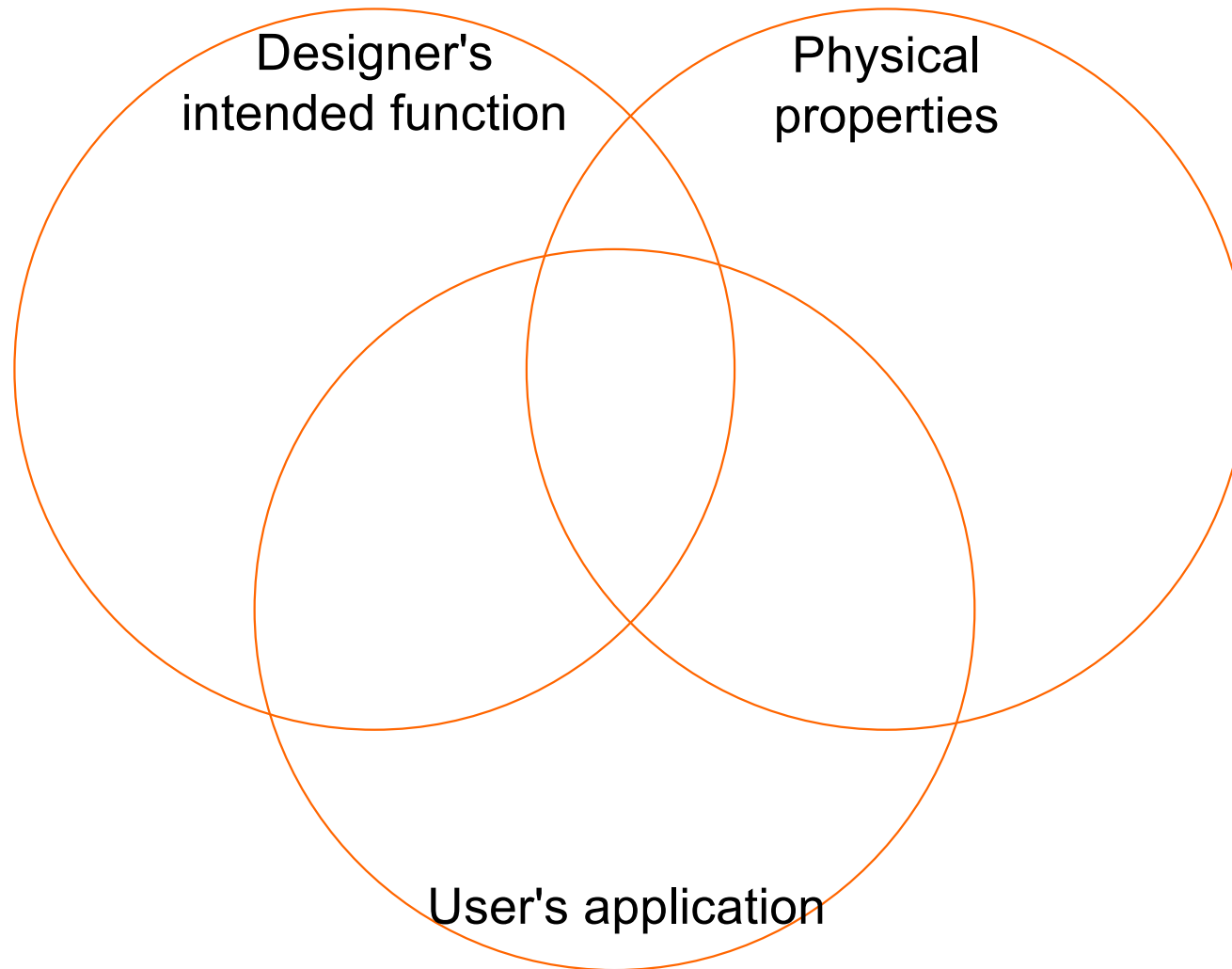
Difficult cases

- Malfunctioning devices
- Failed interventions
- Incidental tools
- Misuse
- Obsolete devices

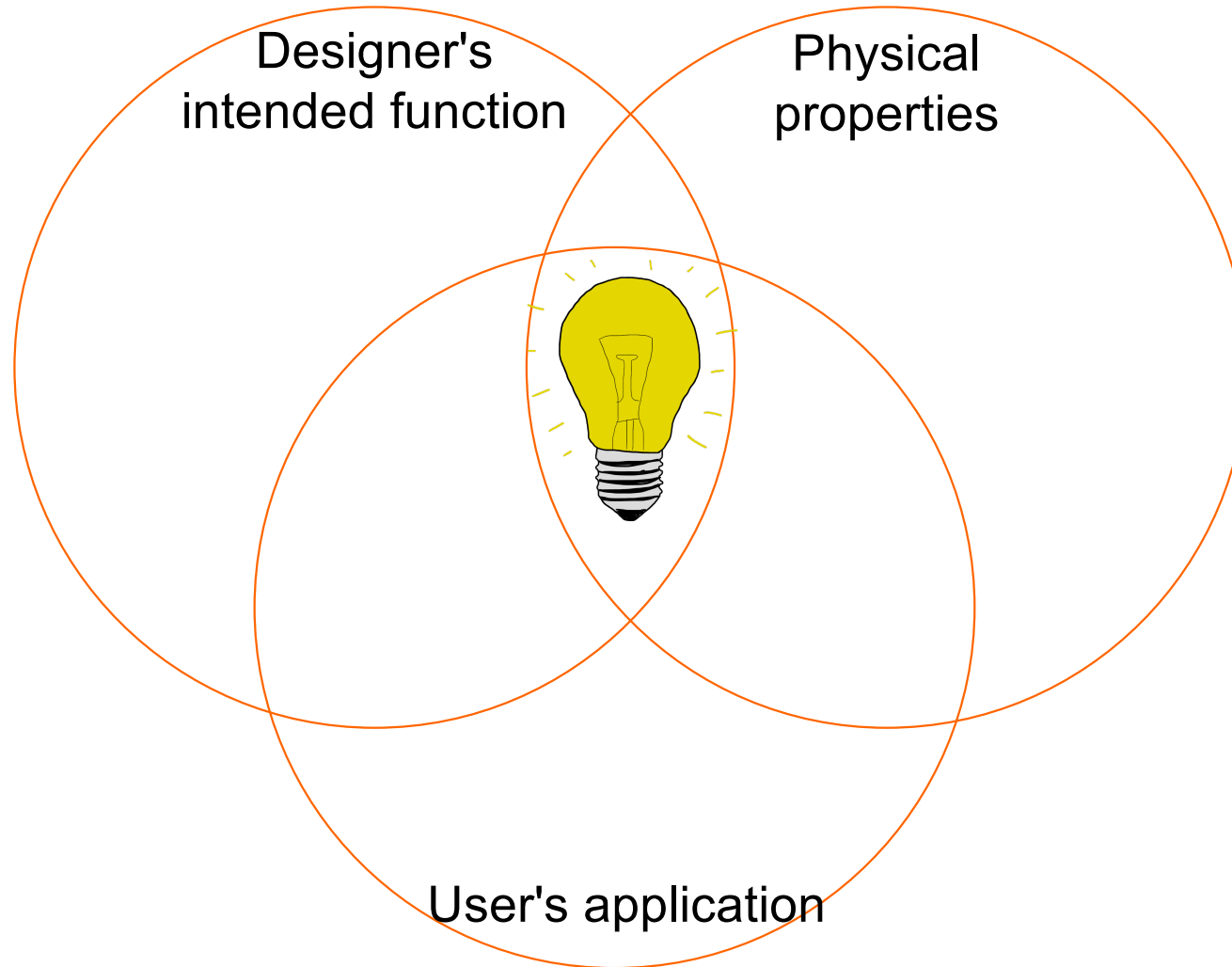


Distinguish designers'
and users' intentions

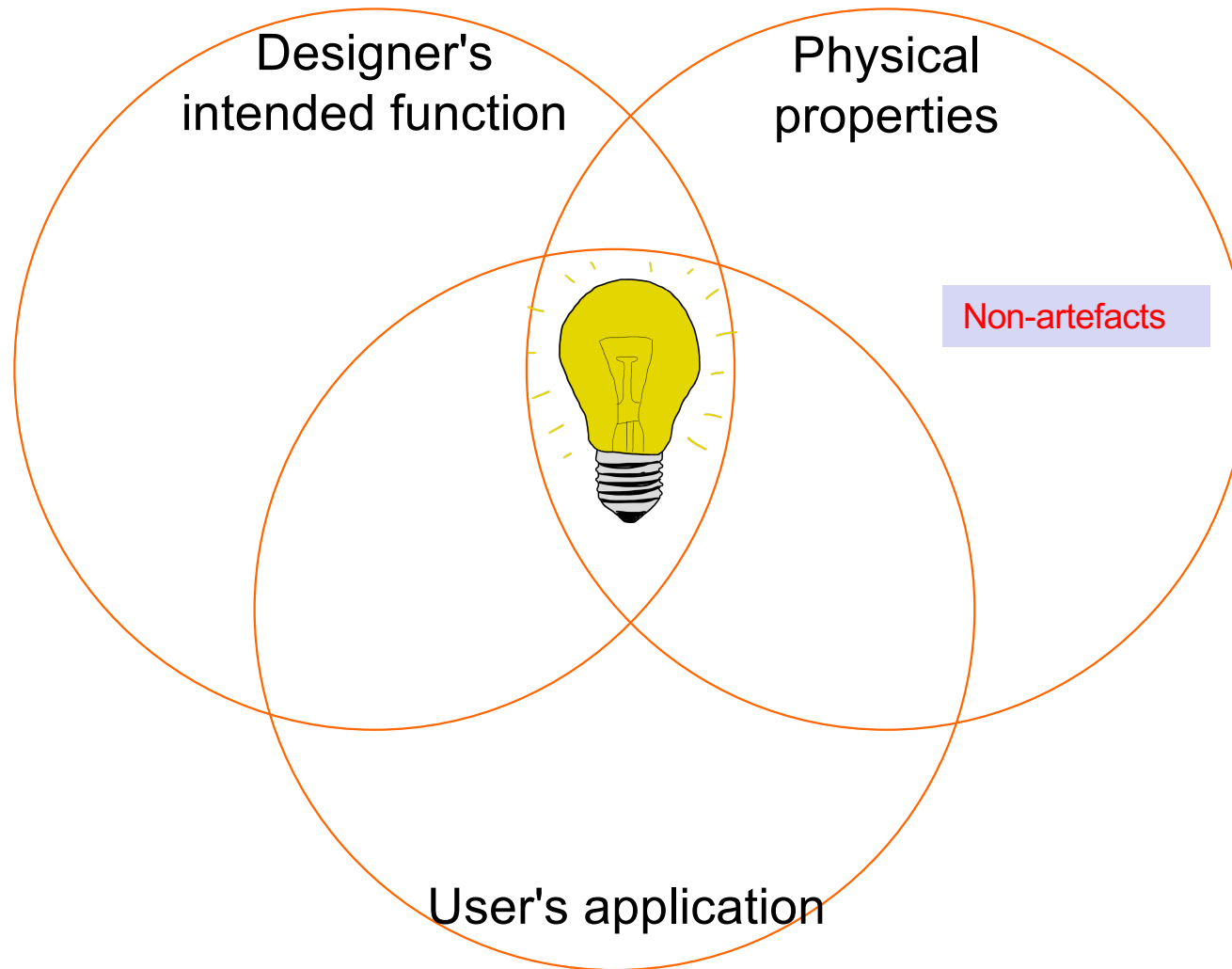
Difficult cases



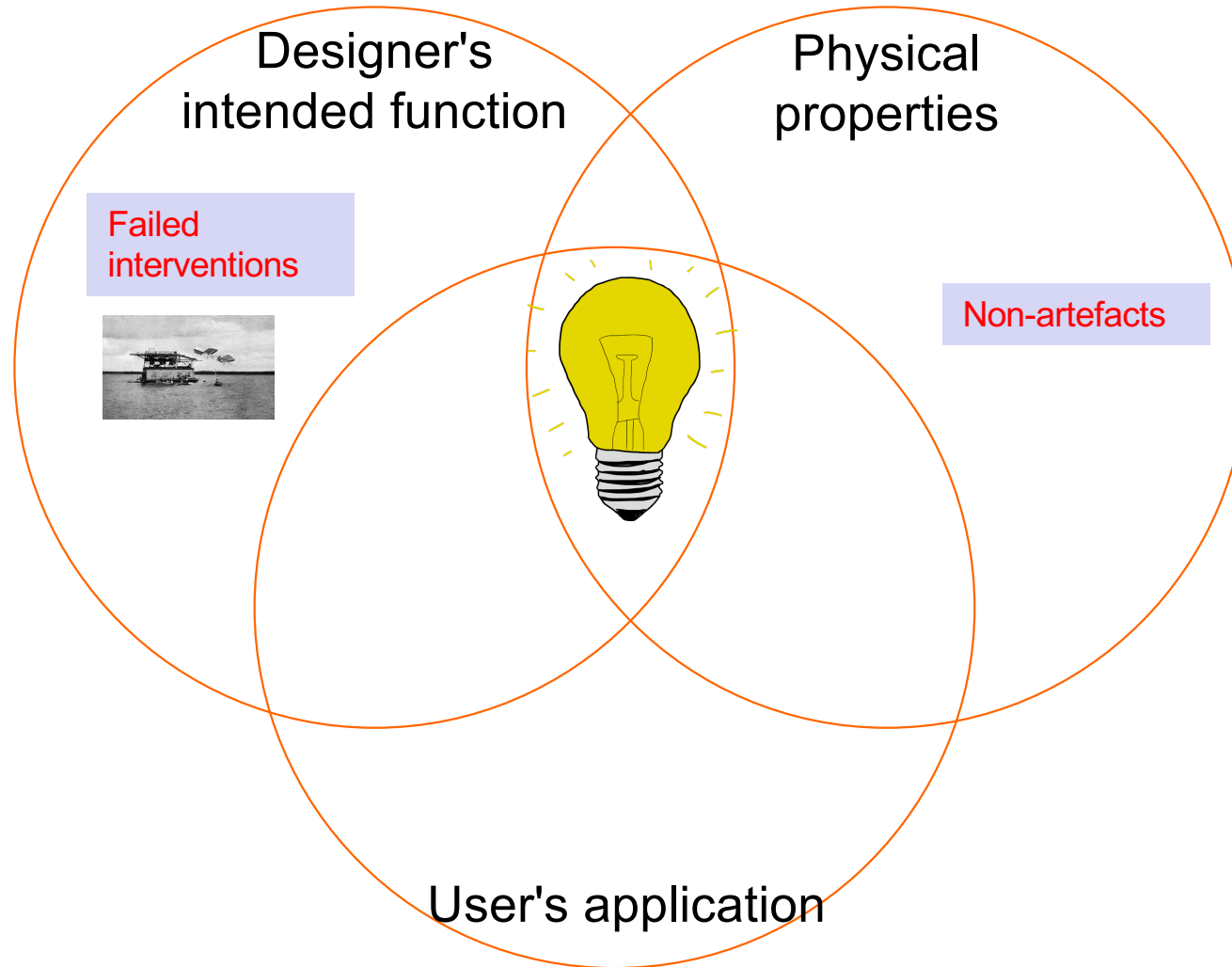
Difficult cases



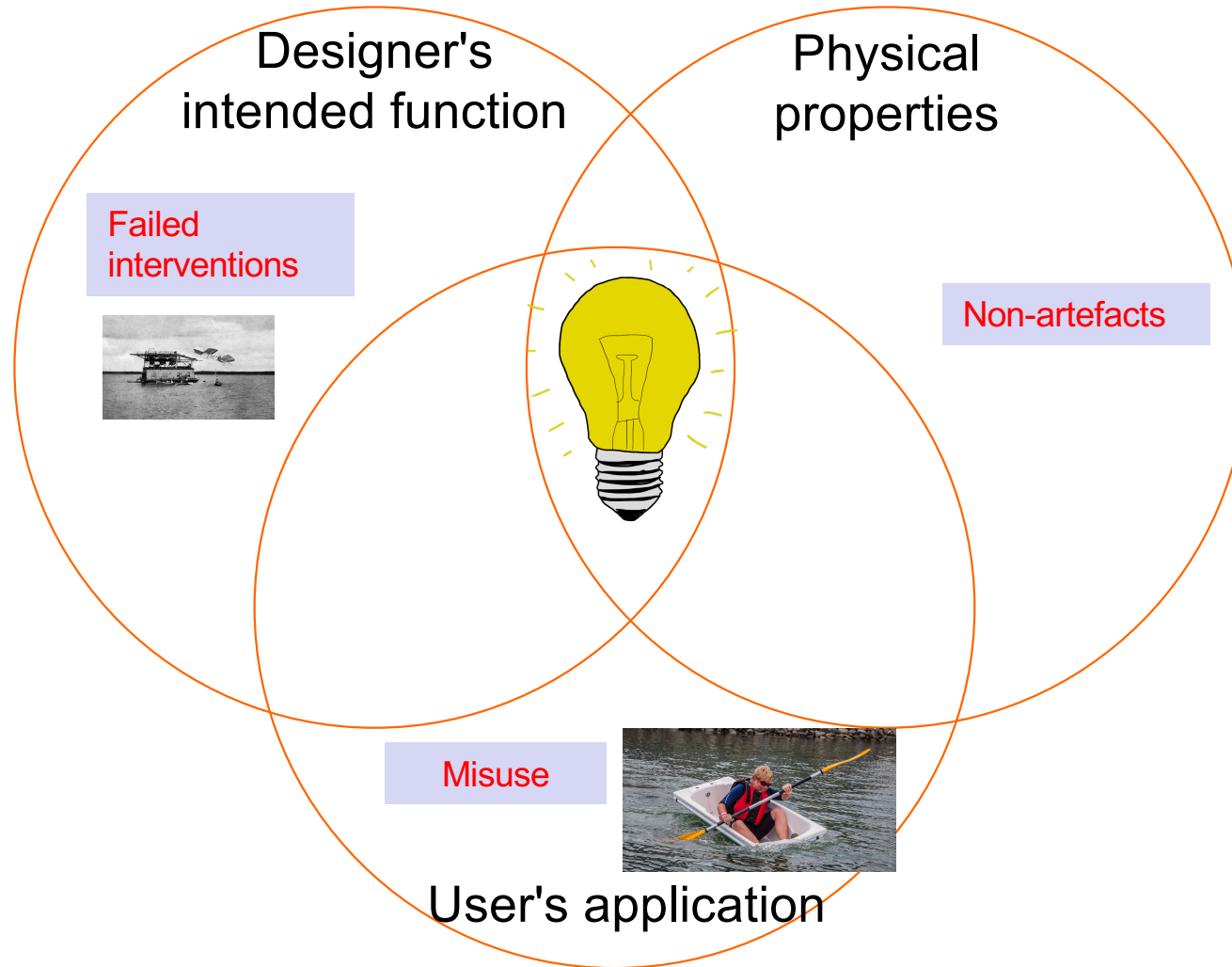
Difficult cases



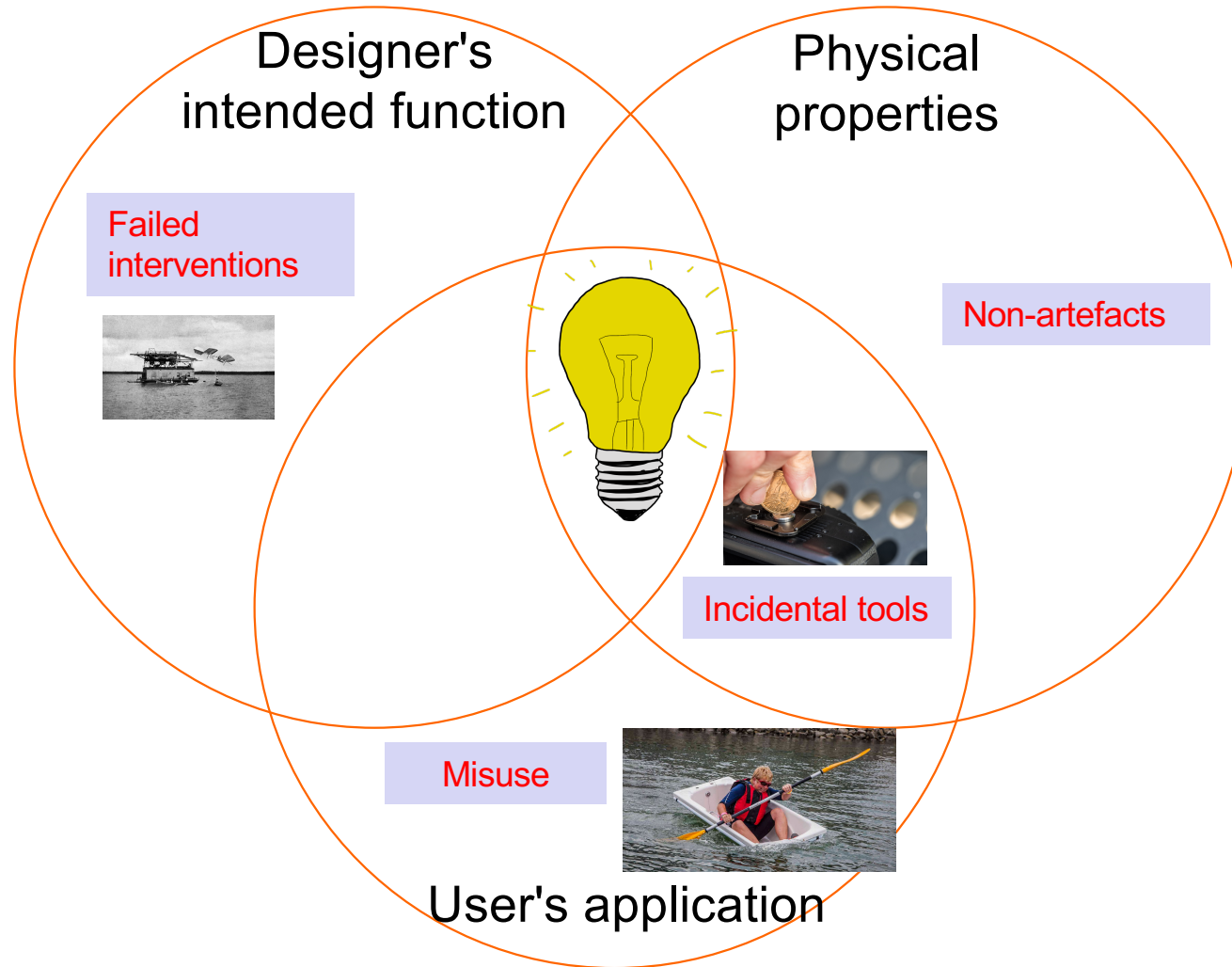
Difficult cases



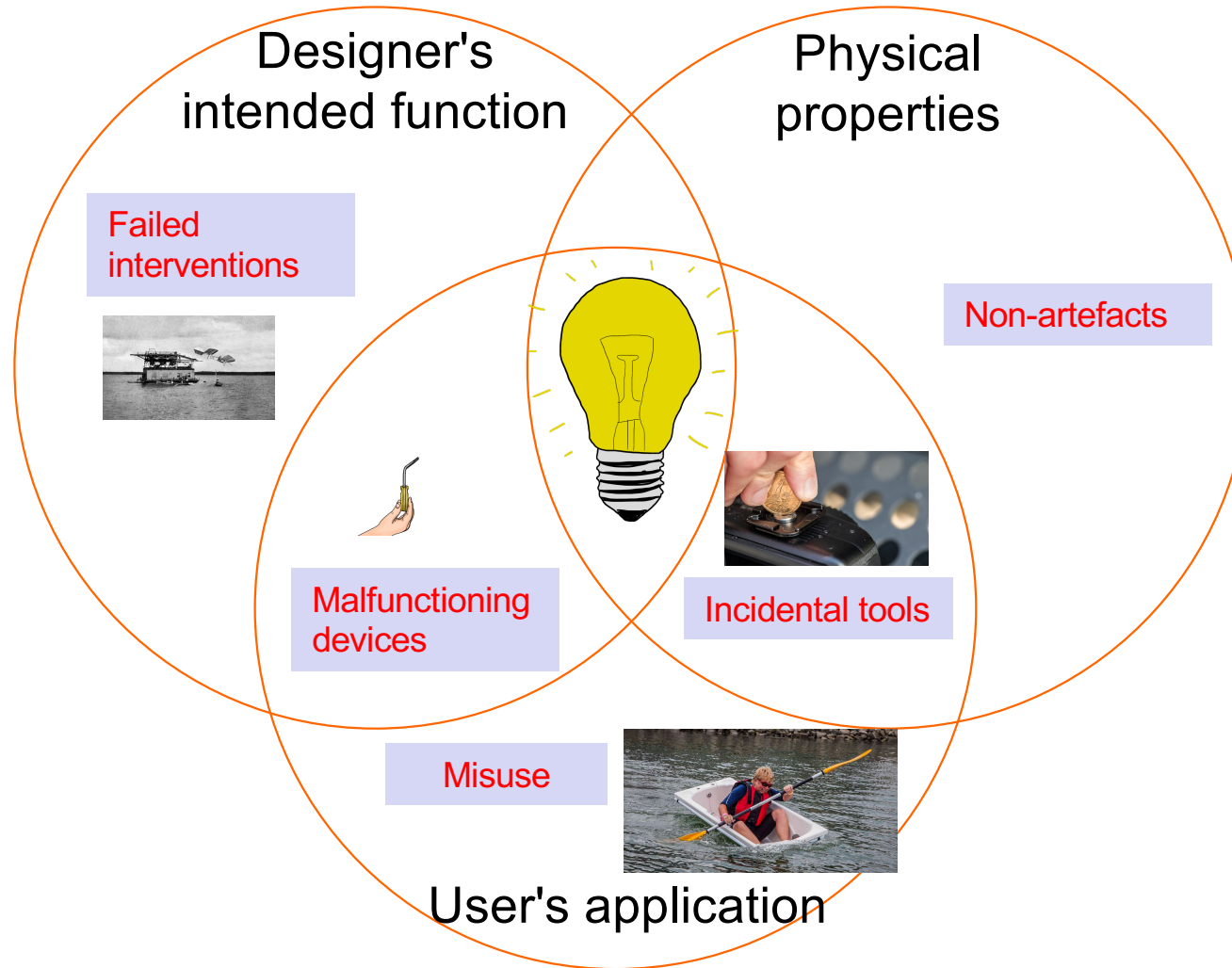
Difficult cases



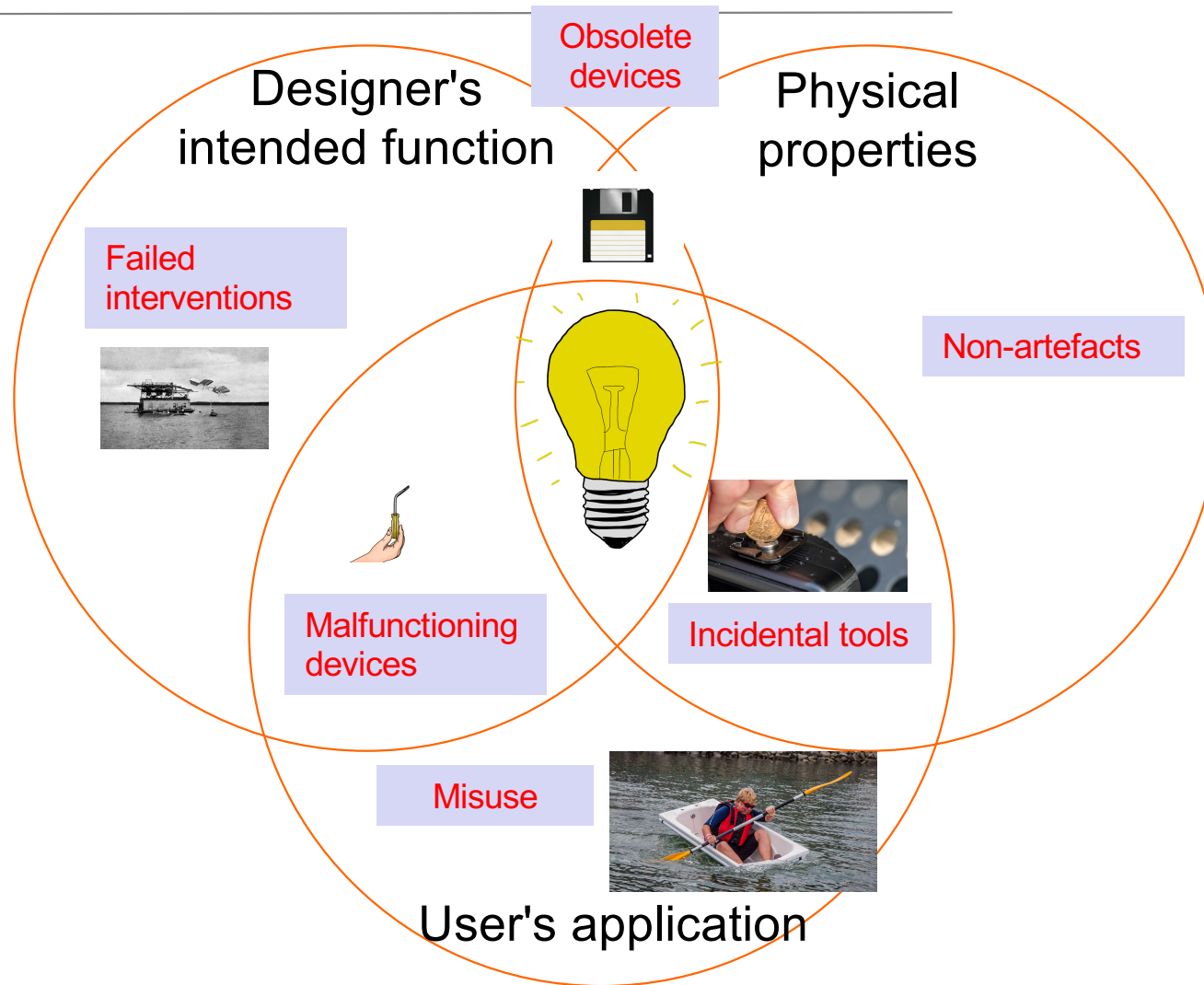
Difficult cases



Difficult cases



Difficult cases



Summary

- Artefacts are material objects that have a function intended by its designers and its users.
- Technical artefacts realize their function in virtue of their physical properties.
- Technical artefacts may malfunction, might have incidental uses, and may become obsolete.

3. Defining technical functions

Functions

Dual account: Technical artefacts are characterized *both* by their **physical properties** *and their intended functions*.

Functions are not just the capacities established by the physical properties.

Instead functions are also established by people's **intentions**: the purposes that someone intend for them. Functions they **ought** to have. -> **Normative quality of artefacts**

Functions & Normativity

Dual account: Technical artefacts are characterized *both* by their **physical properties** and their **intended functions**.

Functions are not just the capacities established by the physical properties.

Instead functions are also established by people's **intentions**: the purposes that someone intend for them. Functions they **ought** to have. -> **Normative quality of artefacts**

“The scientist is concerned with how things are but the engineer with *how things ought to be*”

Herbert Simon 1969 *The sciences of the artificial*

The normative quality of artefacts

Indicated by....

- Artefacts malfunction
- Interventions fail
- Artefacts are misused

Natural scientists, who only focus on physical properties, do not encounter these normative claims. So they must emerge from the functional properties specific to engineering & the artefacts they engage with. How?

Functions & Normativity

But if it's people's intentions with or for objects that establish their functions and thus their normative quality...

two questions are:

- **Whose** intentions?
- **Any** intentions, however crazy?

1st Question: whose intentions?

functions can be established either by the designer or by the user.

In difficult cases, these two can come apart: a user might apply an artefact in a way no designer ever intended it for, and a designer might intend an artefact to function in certain ways without it ever being used this way.

So we can distinguish...

Assigning vs. ascribing functions

Function ascription – a *user* believes that an existing object has certain capacities and therefore *ascribes* it a function

- An epistemic question – belief correct or incorrect in a given context
- No sense of malfunction – if it doesn't work, it's just a matter of a wrong belief
- No sense of artefact misuse – no one is giving guidance but one's own beliefs



Incidental tools

Assigning vs. ascribing functions

Function assignment – a designer *intends* to create an artefact with a certain function, and based on her expertise, describes or creates it.

- A performative act – if competent, designer describes a **new instance of a functional kind**



is of the functional kind that produces visible light from electric current.



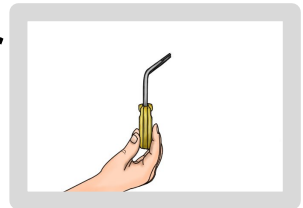
is of the functional kind that flies by gaining support from the air.

Assigning vs. ascribing functions

By identifying an object as an instance of a functional kind, the designer generates a normative claim: this object **ought to** satisfy the function(s) of this kind!

Normative implications:

- **Malfunction:** artefact's physical structure is no longer capable of supporting function of that kind
- **Misuse:** user engages artefact in ways that does not establish function of this kind (and does not establish alternative functions either)

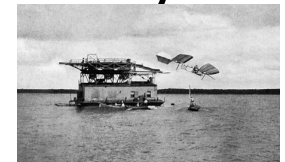


2nd Question: Any intention?

Function assignment – a designer *intends* to create an artefact with a certain function, and based on her expertise, describes or creates it.

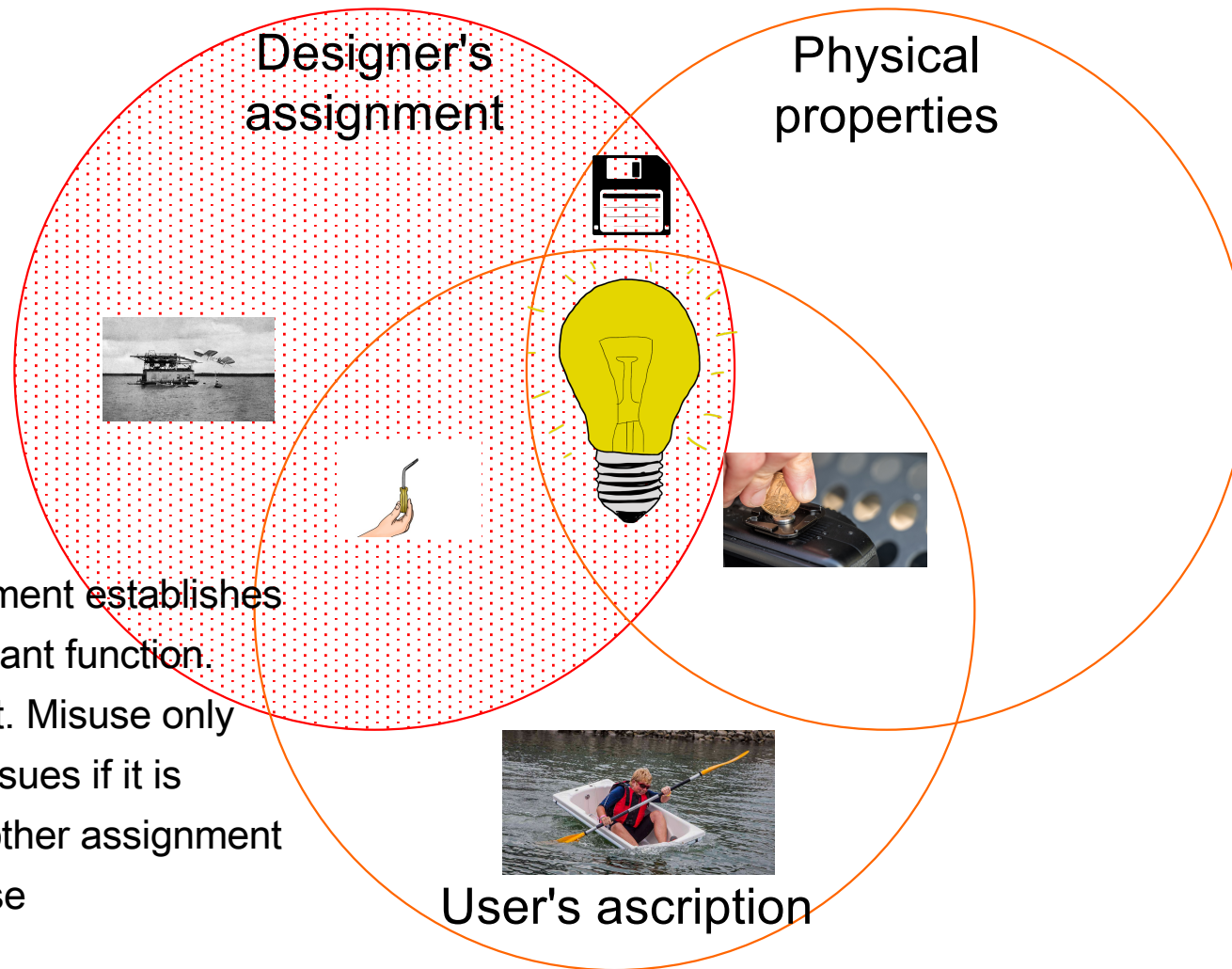
- Failure: designer described/created physical structure not capable of supporting function

Thus, only assignments that are backed by substantially correct expertise (design knowledge) establish function



Langley's aerodrome interesting here because it failed, but was based on a lot of expertise, as e.g. shown through the functioning of the scale model. That's why the Smithsonian treats it as an artefact, even though it failed to function as intended.

The Normative Power of Assignment



Competent assignment establishes a normatively relevant function.

Ascription does not. Misuse only raises normative issues if it is affected by some other assignment different from its use

Summary

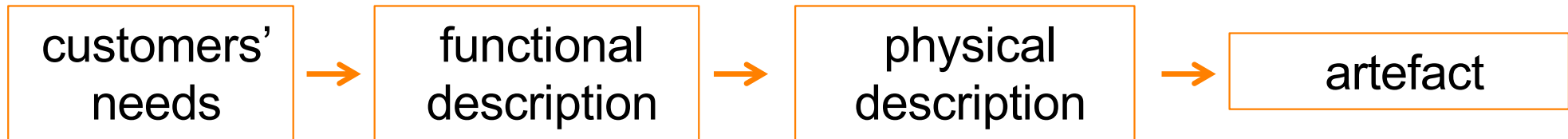
- The functions of an artefact can be established either by designer assignment or user ascription
- Competent designer assignment establishes normative claims about the artefact – i.e. it belonging to a functional type – that allow to determine malfunctions and incidental use.
- Functional ascriptions do not establish normative claims.

4. Knowledge for Design

What is engineering knowledge?

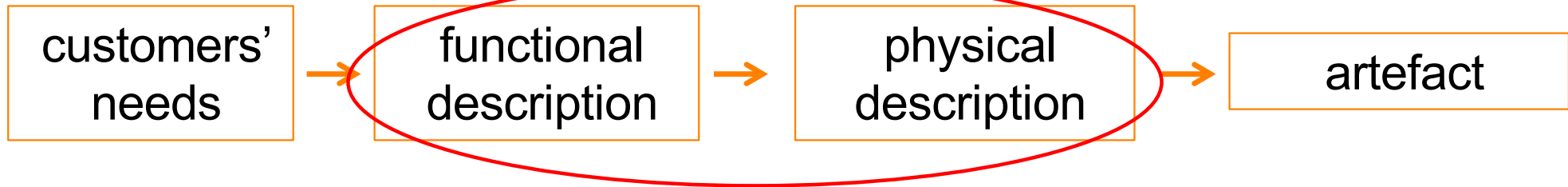
- All artefacts are material objects that operate through cause-effect relations and are governed by laws of nature
- Science knowledge helpful to understand artefacts
 - why does an airplane fly? What produces nuclear power? Why does aspirin reduce headaches?
- But scientific knowledge alone cannot support design, as I will now argue.
- Therefore, let's first have a look at what the design process is that requires this extra knowledge

What is the design process?



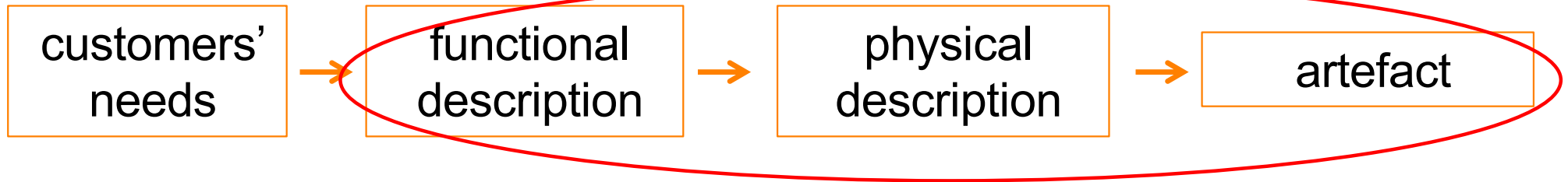
Design generates artefacts by connecting functional and physical descriptions. It creates the dual nature of an object and thus establishes an artefact description.

What is the design process?



Core: A process that develops a physical description corresponding to a functional description

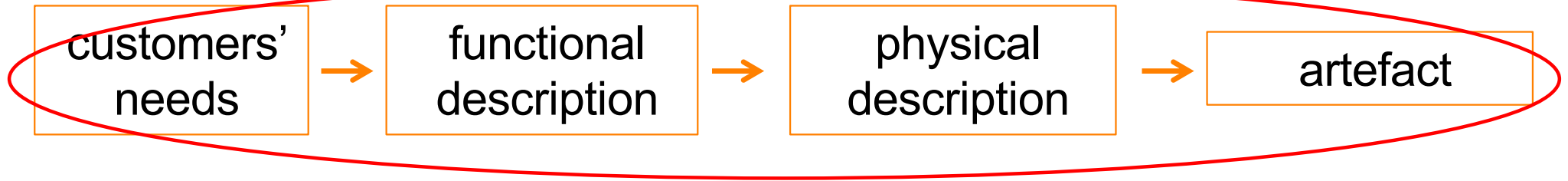
What is the design process?



Core: A process that develops a physical description corresponding to a functional description

Creation-centered: A process that **creates** a physical object that robustly satisfies a functional description

What is the design process?

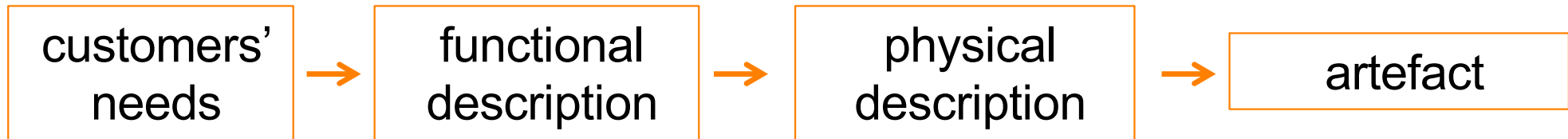


Core: A process that develops a physical description corresponding to a functional description

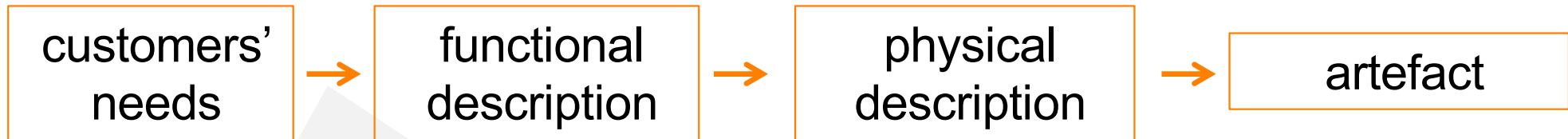
Creation-centered: A process that creates a physical object that robustly satisfies a functional description

Client-oriented: A process that creates a physical object that robustly satisfies the client's needs

What knowledge for the design process?



What knowledge for the design process?



What functions should the device satisfy?

"At present we simply specify that the airplane shall be perfect in all respects and leave it up to the contractor to guess what we really want in terms of degree of stability, controllability, maneuverability, control forces, etc. He does the best he can and then starts building new tails, ailerons, etc. until we say we are satisfied" (Capt. RS Hatcher, Navy Bureau of Aeronautics, April 1941).

- Develop functional concepts
- Learn what clients/users want

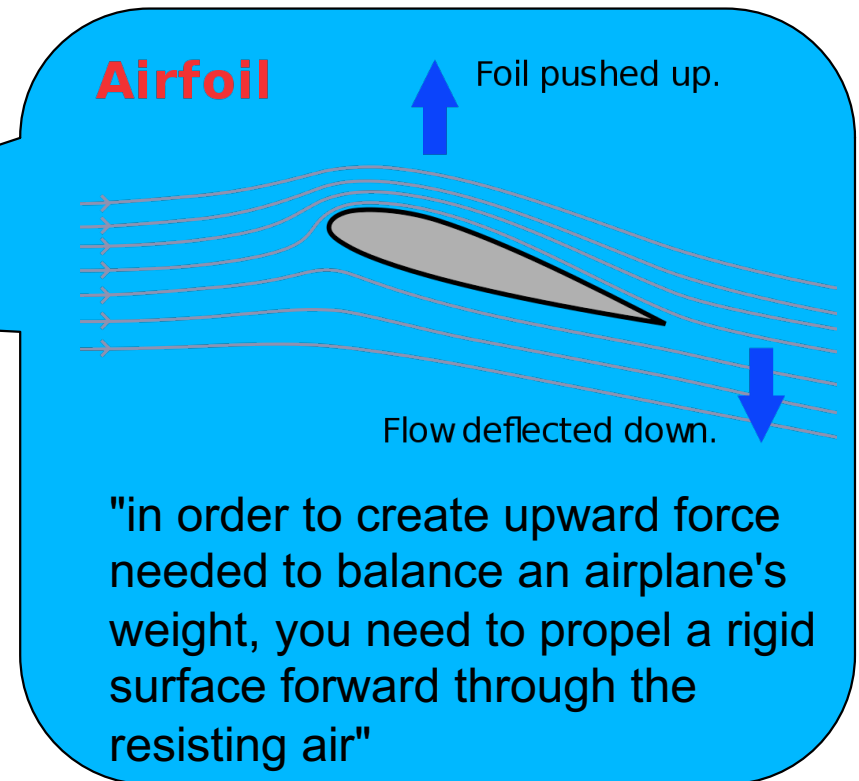
What knowledge for the design process?



Which physical structures might satisfy the demanded functions?

Theory

- Operational principles



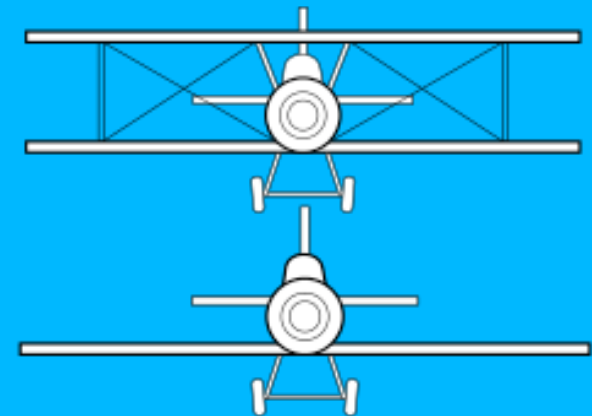
What knowledge for the design process?



Which physical structures might satisfy the demanded functions?

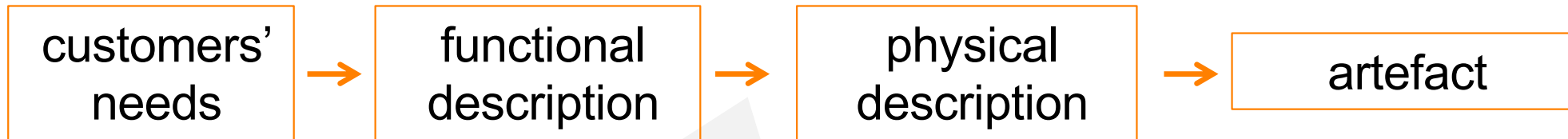
Theory

- Operational principles
- Normal configurations



the general shape and arrangement commonly agreed to best realize the operational principle

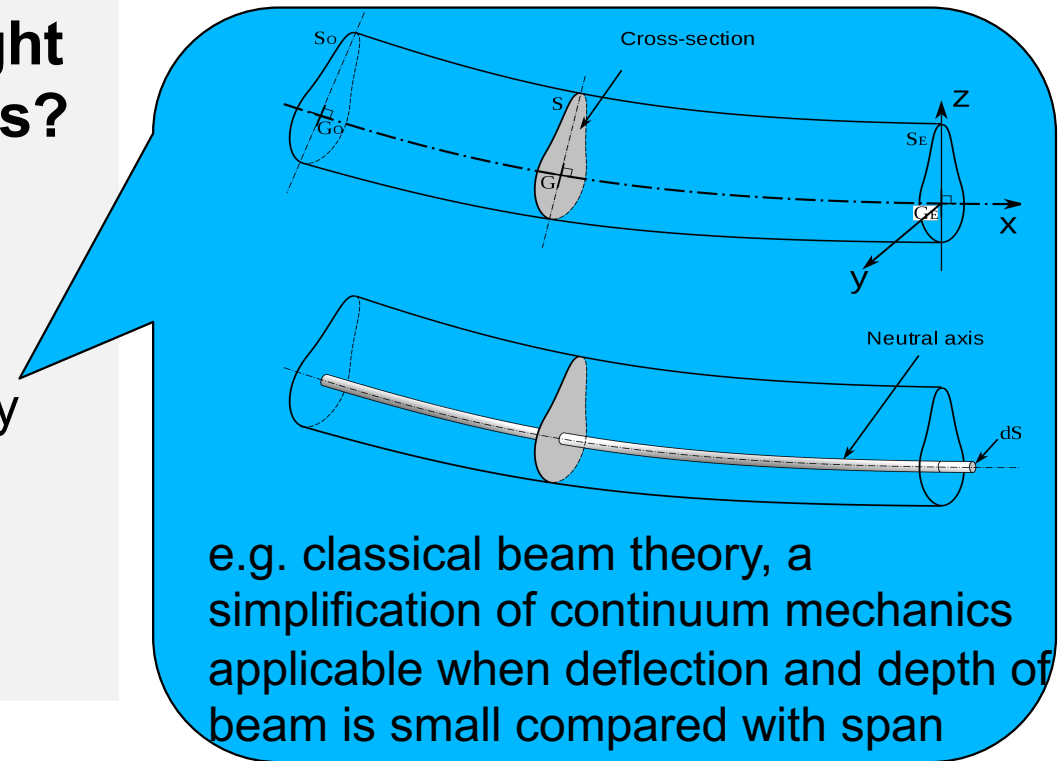
What knowledge for the design process?



Which physical structures might satisfy the demanded functions?

Theory

- Operational principles
- Normal configurations
- Devise-restricted simplifying theory



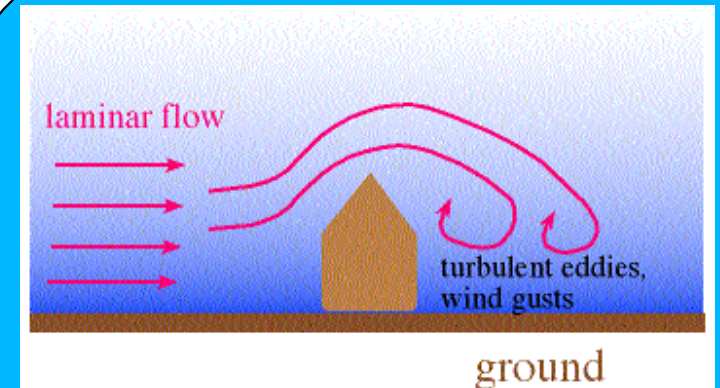
What knowledge for the design process?



Which physical structures might satisfy the demanded functions?

Theory

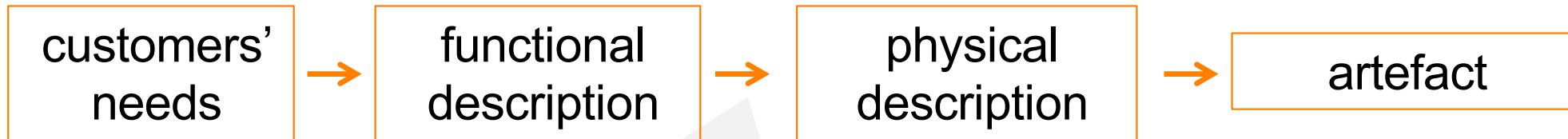
- Operational principles
- Normal configurations
- Devise-restricted simplifying theory
- Phenomenal theory



Eddy viscosity. Ignore the small-scale vortices (or eddies) in the motion. Calculate a large-scale motion with an effective viscosity.

- Not physically rigorous
- Might be faulty

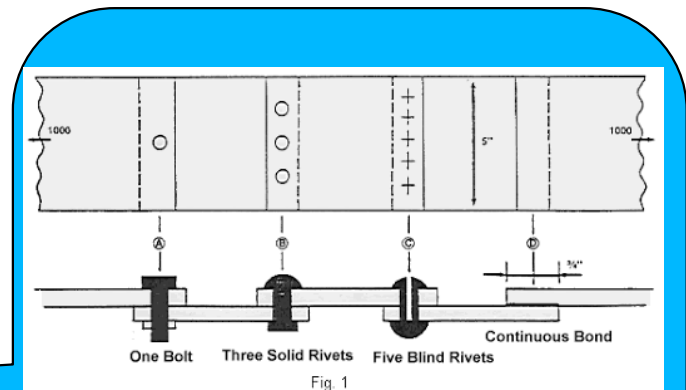
What knowledge for the design process?



Which physical structures might satisfy the demanded functions?

Theory

- Operational principles
- Normal configurations
- Devise-restricted simplifying theory
- Phenomenal theory
- Quantitative assumptions



e.g. riveted joints analysis:
assume loads on joint equally
shared between all rivets

What knowledge for the design process?

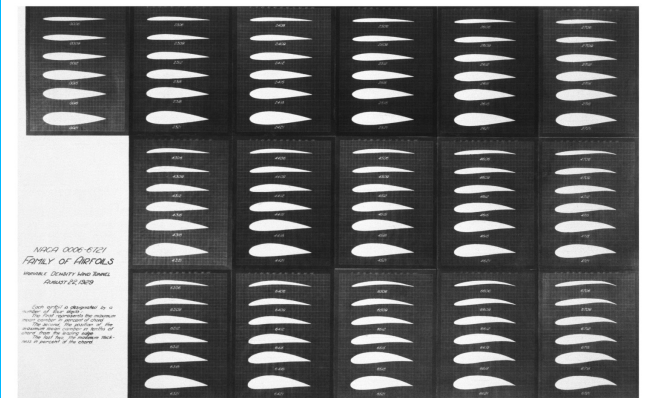


Which physical structures might satisfy the demanded functions?

Theory

- Operational principles
- Normal configurations
- Devise-restricted simplifying theory
- Phenomenal theory
- Quantitative assumptions

Quantitative data

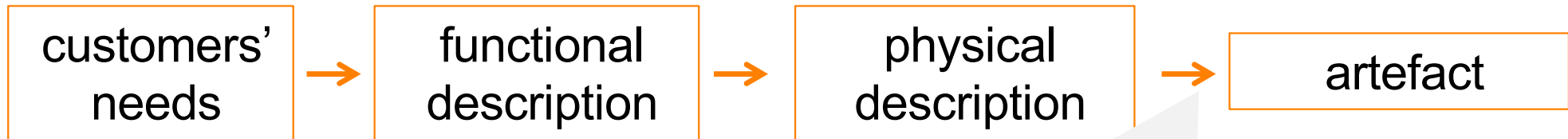


NACA L-5344

Langley Research Center
Hampton, Virginia 23615-0001

e.g. NACA airfoil catalogue with their respective windtunnel performances (1929)

What knowledge for the design process?



e.g. how much clearance for assembling worker to reach inside?

How to produce these physical structures?

- Material, labour, expertise availability
- Efficiency considerations
- Scaling problems
- Practical considerations

Science vs. Design Knowledge

Engineering	Science
Investigate functional demands	No concept of technical function
work with very concrete functional objectives	few rigidly specified goals beyond ultimate goal of understanding.
Organize theory & data around functional kinds	Organize theory & data around natural kinds
Allow for device- or context-specific knowledge, where convenient	Always aim for generality
if the devices to which they apply cease to be useful, the knowledge will not longer be of interest	Once acquired knowledge always of interest

"The scientific knowledge of a machine as an object tells us nothing about it as a machine."

Michael Polanyi 1958

Summary

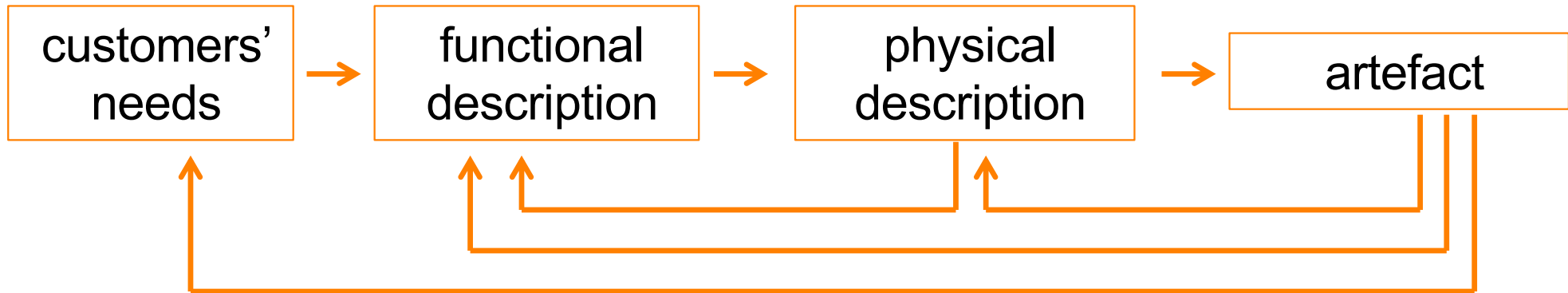
- Design, at its core, is the translation of a functional into a satisfactory physical description
- Of course, engineers often draw on scientific knowledge, but it is neither enough nor of the right kind.
- But scientific knowledge alone cannot support design
- Other kinds of knowledge are necessary: functional demands, engineering theory, engineering data, practical considerations
- Engineering design requires its won knowledge – it is not just “applied science”

5. Design Methodology

Is there a methodology of design?

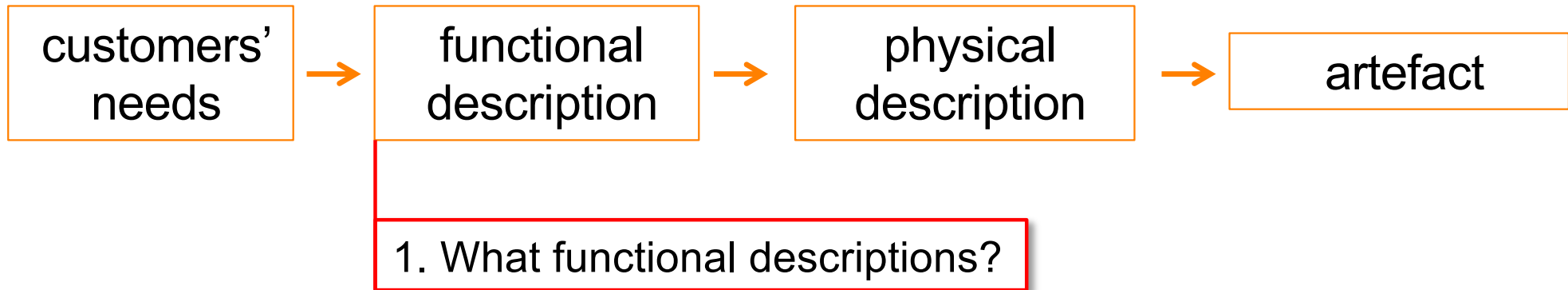
- Design requires special kinds of knowledge
- There are many methods that could generate and apply this knowledge
- What justifies this method choice?
- -> Methodology of design

Is there a methodology of design?

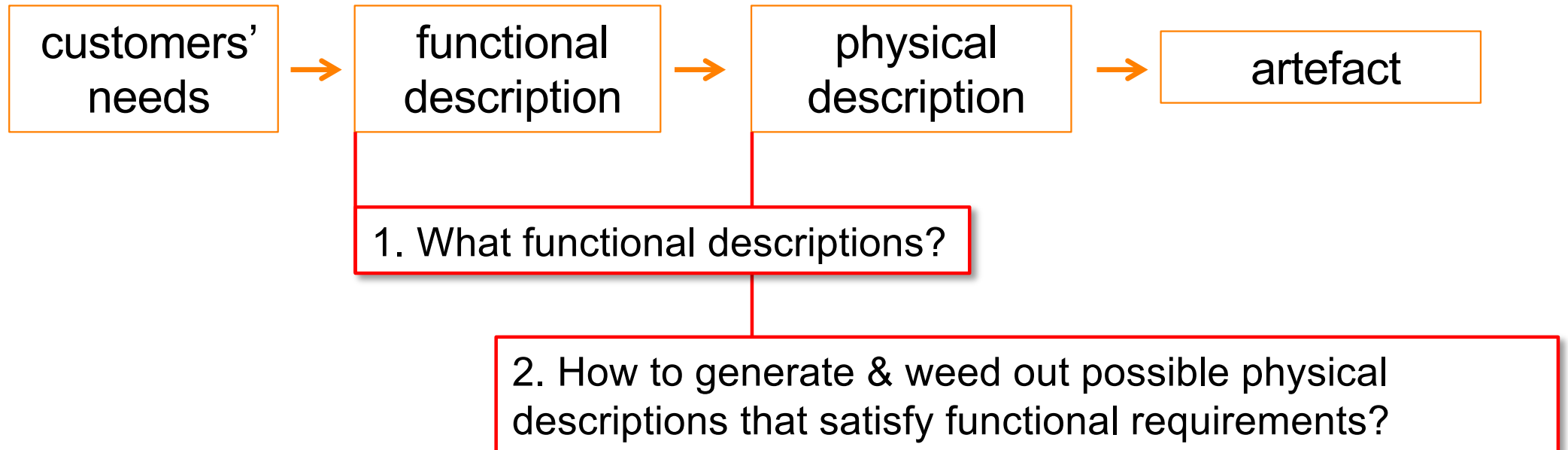


- The simple model of design I presented before gives a sequence of steps that the designer proceeds through (presumably with many iterations)
- At different steps, different method choices must be made

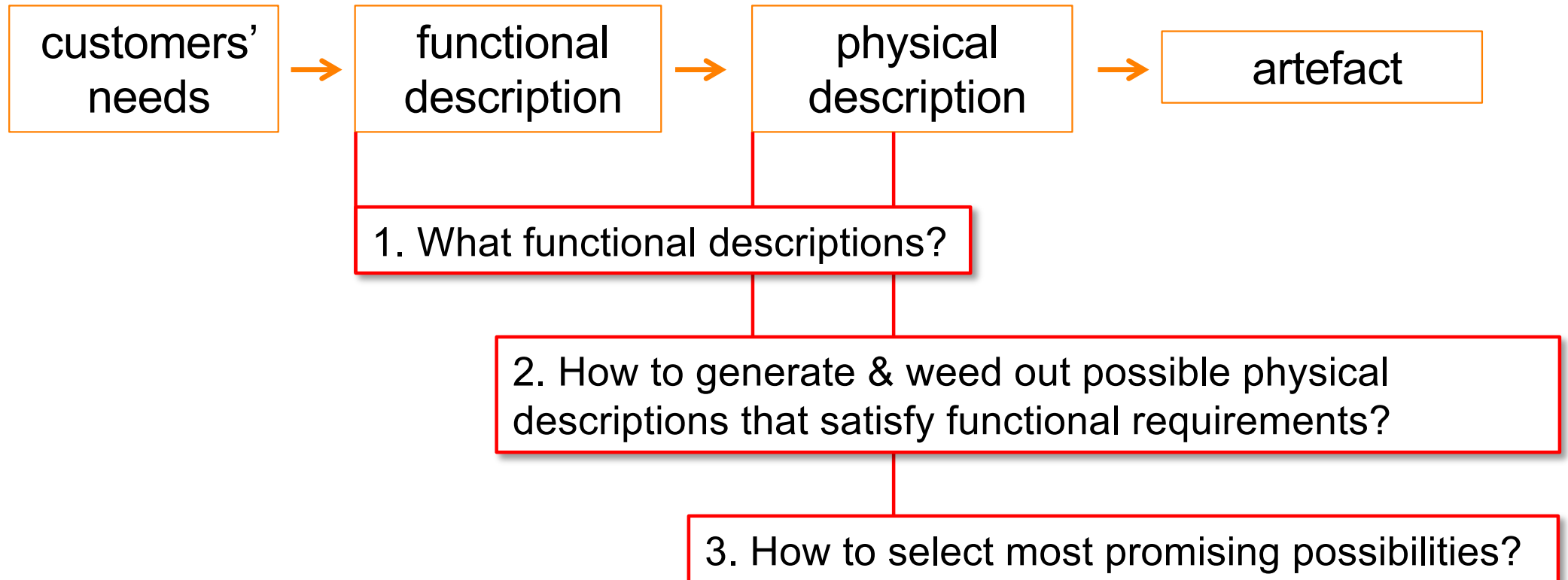
Is there a methodology of design?



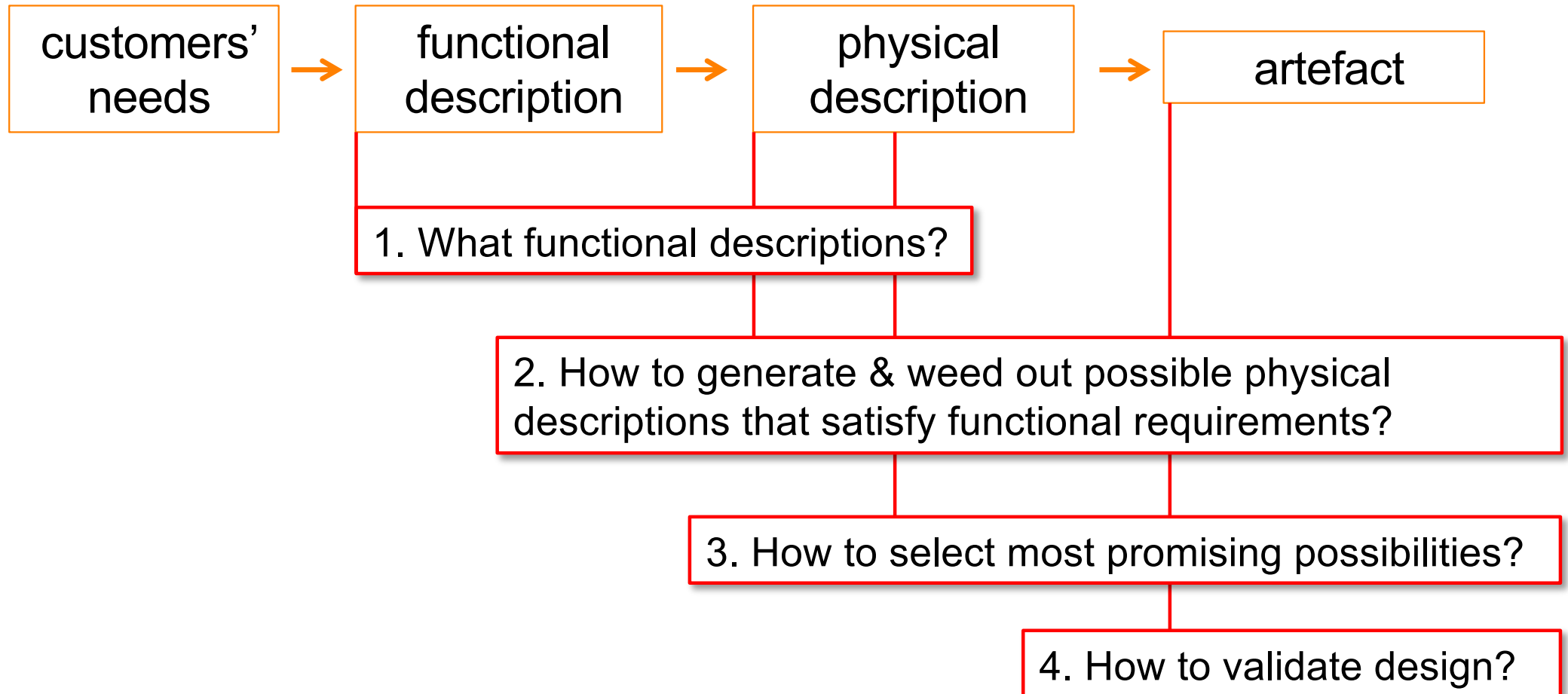
Is there a methodology of design?



Is there a methodology of design?



Is there a methodology of design?



1. Appropriate functional requirements?

Completeness: describe *all* desired effects of the artifact

Use-context: make functional descriptions dependent on users' behavior

Quantitative criteria: specify thresholds of failure and degrees of success

Implementation-independence: avoid specifications of physical implementation (i.e. **what it's for, not what it is made of!**)

2. Generate set of physical descriptions

Functional decomposition

Split functions of artefact into sub-functions

At each level of decomposition, check whether there are standard solutions for function

- operational principles

- normal configurations

Reiterate!

Blind variation

Sub-functions for which no standard solution can be found must be given a physical description without clear knowledge base – in that sense “blind”

“blind” does not mean randomly or unconstrained. lot of knowledge, both theoretical and empirical is about what DOESN'T work, and this can act as a constraint

Analogies to other artefacts or models help specify details

Variation within these constraints is important – “sweeping the reasonable parameter space”

3. Decide between possibilities

Optimization

- Exhaustive search -> complete set of design alternatives
- For each alternative, determine how *probable* it satisfies functions
- Calculate expected value for each design: probability-weight sum of the value of its functions
- Determine best design by maximizing expected value

3. Decide between possibilities

Optimization

Satisficing (term introduced by H Simon in 1950s, combines **satisfy** and **suffice**)

- Determine minimal functional threshold: which functions must be satisfied? Which functions might be traded off against each other?
- Begin search
- Evaluate each design: does it meet threshold criteria?
- If yes, end search.
- If no, continue.
- (If search is futile, adjust threshold)

4. Design validation

Testing aimed at ensuring that a product or system fulfills the specified functional requirements, under specified operating conditions.

- Proof of concept
- Prototype
- Direct trial
- Usability testing

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Testing aimed at ensuring that a product or system fulfills the specified functional requirements, under specified operating conditions.

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Performed through material modelling, experimenting, observational studies, etc. Similar criteria as in science apply on the lower levels. But the goal is different: they don't aim to show what is true, but whether object satisfies functional requirements

Summary

- Even the simple sequential model of design identifies various points of method choice.
- Little agreement on how to justify these method choices in general
- Identified some criteria that might be relevant
- Different method choices and different criteria than in science